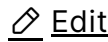


<https://hackmd.io/@aligho/r1zDQZNgbg/edit?both> / CP...[Edit both](https://hackmd.io/@aligho/r1zDQZNgbg/edit?both)

CPA bandwidth for spherical systems

In this set of notes, we try to find an analytic approximation for the CPA bandwidth corresponding to coated spherical systems.

From <https://hackmd.io/@aligho/HyiKTnQkbe>

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$$s \approx \frac{2 \left[(1 - \epsilon_m) - 2g(\omega) \right] x_2^3 + 3i \left[(2 + \epsilon_m) + 2g(\omega) \right]}{2 \left[(1 - \epsilon_m) - 2g(\omega) \right] x_2^3 - 3i \left[(2 + \epsilon_m) + 2g(\omega) \right]}$$

The CPA condition at a given frequency, $\omega = \omega_0$, will be satisfied if we have $g(\omega) \approx -(2 + \epsilon_m)/2 + i(a\omega_0/c)^3$, we have that the scattering coefficient's behavior near the CPA point will be given by (for small x_2):

$$|s|^2 \approx 1 - \frac{4(a\omega_0/c)^6}{4(a\omega_0/c)^6 + [\Re(g_1)'(\omega_0)(\omega - \omega_0)]^2}$$

In the expression above, we ignored derivatives of the imaginary part of $g(\omega)$ since this is small compared to those of the real part for $x_2 < 1$ (and, correspondingly $\omega\tau \gg 1$).

From the above, we may read off the half width at half maximum, Γ as:

$$\Gamma \approx \frac{2(a\omega_0/c)^3}{|g'(\omega_0)|}$$

We now use the fact that $g_1(\omega) \propto 1/(\omega^2 + 1/\tau^2)$ and furthermore use the fact that we are in the small sphere regime where $\tau\omega \gg 1$ in order to write $|g'_1(\omega_0)| \approx (2 + \varepsilon_m)/\omega_0$

This then gives us:

$$\Gamma \approx \frac{2(a\omega_0/c)^3\omega_0}{(2 + \varepsilon_m)}$$

We note that $\omega_0\tau = \Im\sigma/\Re\sigma$, which means that we have that $\omega_0 \propto (2 + \varepsilon_m)/[2(a\omega_0/c)^3]$

Therefore, we have that Γ is almost constant regardless of the CPA frequency we consider.